KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

NAME:

TEACHER:

MATHEMATICS

Year 8 – Practice Examination 1

Equations, Investigating Data and Mixed Topics

Practice paper for the Year 8 Yearly Examination

Instructions		
1. Attempt all questions 2. Show working where needed 3. Time Allowed 60 minutes 4. Calculators are permitted 5. Marks are shown beside each question	Equations (Section A):	/18
	Investigating Data (Section B):	/21
	Mixed Topics (Section C):	/21
	Total Marks:	/60

Section A – Equations (18 marks)

Suggested time: 20 minutes. Show all working.

Question 1 – One-step equations **(4 marks)**

Solve each equation.

a) $x + 8 = 21$ x = _____ **(1 mark)**

Working: _____

b) $m - 12 = 5$ m = _____ **(1 mark)**

Working: _____

c) $7p = 56$ p = _____ **(1 mark)**

Working: _____

d) $y \div 4 = 9$ y = _____ **(1 mark)**

Working: _____

Question 2 – Two-step equations **(4 marks)**

Solve each equation, showing your working.

a) $5x + 3 = 38$ **(2 marks)**

Working: _____

_____

_____

x = _____

b) $x \div 6 - 1 = 4$ **(2 marks)**

Working: _____

_____

_____

x = _____

Question 3 – Equations with a negative solution **(2 marks)**

Solve $3m + 14 = 2$.

Working: _____

_____

_____

m = _____

Question 4 – Checking solutions **(2 marks)**

Use substitution to show whether $x = 6$ is the solution to $4x - 9 = 13$.

Working: _____

_____

_____

_____

Is x = 6 the solution? (circle one) YES / NO

Question 5 – Worded problem **(3 marks)**

I think of a number, multiply it by 3, and then add 8. The answer is 35.

a) Write an equation to represent this problem. Use *n* for the number. **(1 mark)**

Equation: _____

b) Solve your equation to find the number. **(2 marks)**

Working: _____

_____

_____

n = _____

Question 6 – Worded problem **(3 marks)**

Liam buys 4 movie tickets that each cost the same amount and a popcorn that costs $7. He spends $55 altogether. Let *t* be the cost of one movie ticket in dollars.

a) Write an equation to represent this situation. **(1 mark)**

Equation: _____

b) Solve your equation to find the cost of one movie ticket. **(2 marks)**

Working: _____

_____

_____

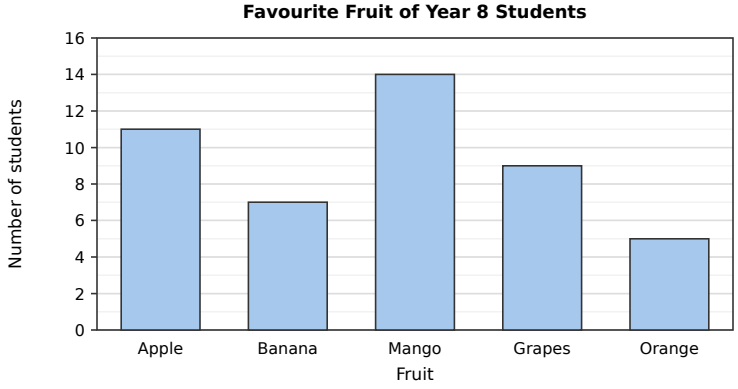
t = _____

Section B – Investigating Data (21 marks)

Suggested time: 20 minutes. Show all working.

Question 7 – Column graph **(2 marks)**

Year 8 students were asked to name their favourite fruit. The results are shown below.



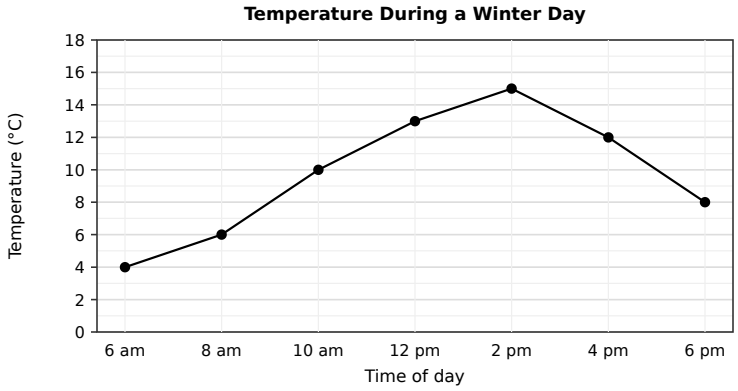
a) Which fruit was the most popular? _____ **(1 mark)**

b) How many students were surveyed altogether? _____ **(1 mark)**

Working: _____

Question 8 – Line graph **(2 marks)**

The line graph shows the temperature at different times on a winter day.



a) At what time was the temperature the highest? _____ **(1 mark)**

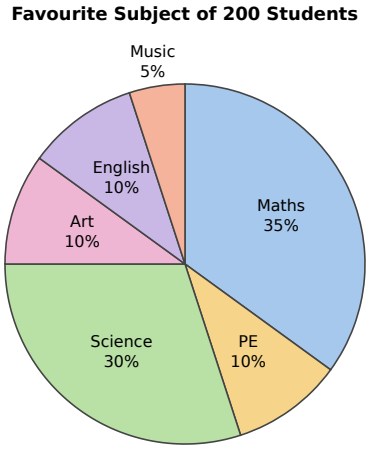
b) By how many degrees did the temperature rise between 8 am and 2 pm? **(1 mark)**

Working: _____

Answer: _____

Question 9 – Sector (pie) graph **(3 marks)**

200 students were asked to name their favourite subject. The results are shown in the sector graph.



a) How many students chose Science? **(1 mark)**

Working: _____

Answer: _____

b) How many more students chose Maths than Art? **(2 marks)**

Working: _____

_____

_____

Answer: _____

Question 10 – Stem-and-leaf plot **(2 marks)**

The number of points scored by 13 basketball players over a season is shown in the stem-and-leaf plot.

Stem	Leaf
1	2 5 8
2	0 3 3 7
3	1 4 6 9
4	2 5

Key: 1 | 2 means 12

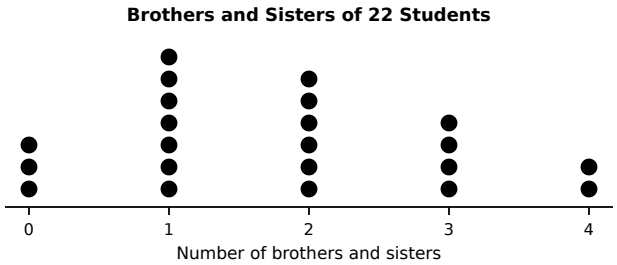
a) Find the median. _____ **(1 mark)**

Working: _____

b) How many players scored more than 30 points? _____ **(1 mark)**

Question 11 – Dot plot **(2 marks)**

Twenty-two students were asked how many brothers and sisters they have. The results are shown in the dot plot.



a) What is the mode? _____ **(1 mark)**

b) How many students have at least 3 brothers and sisters? _____ **(1 mark)**

Question 12 – Frequency table **(2 marks)**

Year 8 students were asked how many hours of sport they play each week. The results are shown in the frequency table.

Hours of sport	0	1	2	3	4	5
Frequency	3	6	8	11	7	3

a) How many students were surveyed altogether? _____ **(1 mark)**

Working: _____

b) What is the mode? _____ **(1 mark)**

Question 13 – Mean, median, mode and range **(5 marks)**

The number of minutes that eight students took to walk to school is shown below.

6, 13, 9, 15, 9, 11, 17, 8

a) Calculate the mean. **(2 marks)**

Working: _____

_____

_____

Mean = _____

b) Find the median. _____ **(1 mark)**

Working: _____

c) Find the mode. _____ **(1 mark)**

d) Find the range. _____ **(1 mark)**

Working: _____

Question 14 – Outliers **(3 marks)**

The number of minutes that six students waited for the bus is shown below.

6, 8, 7, 5, 34, 6

a) Identify the outlier in this data set. _____ **(1 mark)**

b) Calculate the mean of the data, including the outlier. **(1 mark)**

Working: _____

_____

Mean = _____

c) Does the outlier make the mean higher or lower than most of the data values? Explain your answer. **(1 mark)**

Working: _____

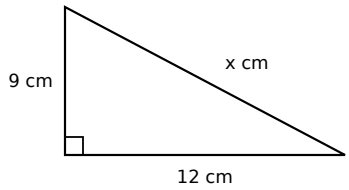
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Section C – Mixed Topics (21 marks)

Suggested time: 20 minutes. Show all working.

Question 15 – Pythagoras' Theorem **(4 marks)**

a) Find the length of the hypotenuse, *x*, in the triangle below. **(2 marks)**



Working: _____

_____

_____

x = _____

b) A triangle has side lengths 10 cm, 24 cm and 26 cm. Is it a right-angled triangle? Show working. **(2 marks)**

Working: _____

_____

_____

_____

Question 16 – Decimals **(2 marks)**

a) Calculate 3.6 × 0.25 _____ **(1 mark)**

Working: _____

_____

b) Calculate 5.4 ÷ 0.09 _____ **(1 mark)**

Working: _____

_____

Question 17 – Powers and roots **(2 marks)**

a) Evaluate 5^3 _____ **(1 mark)**

Working: _____

b) Evaluate √121 _____ **(1 mark)**

Working: _____

Question 18 – Surds **(3 marks)**

a) Circle the surds: √16 √18 √36 √7 **(1 mark)**

b) Simplify √48 **(2 marks)**

Working: _____

_____

_____

Answer: _____

Question 19 – Algebra **(3 marks)**

a) Simplify $6x + 2y - 4x + 7y$ _____ **(1 mark)**

Working: _____

b) Expand $5(a + 3)$ _____ **(1 mark)**

Working: _____

c) Evaluate $3p^2 - 2q$ when $p = 5$ and $q = 6$. _____ **(1 mark)**

Working: _____

_____

Question 20 – Area and volume **(3 marks)**

a) Find the area of a trapezium with parallel sides 6 cm and 10 cm and height 7 cm. **(2 marks)**

Formula: Area = ½ × (a + b) × h, where a and b are the parallel sides and h is the perpendicular height.

Working: _____

_____

_____

Answer: _____

b) Find the volume of a rectangular prism that is 8 cm long, 5 cm wide and 2 cm high. **(1 mark)**

Working: _____

_____

Answer: _____

Question 21 – Linear relationships **(2 marks)**

Complete the table of values for the rule $y = 3x + 1$.

x	−2	−1	0	1
y				

Question 22 – Rates and ratios **(2 marks)**

a) Write the ratio 15 : 25 in simplest form. _____ **(1 mark)**

Working: _____

_____

b) A cyclist rides 72 km in 4 hours. Find the cyclist's average speed in km/h. **(1 mark)**

Working: _____

_____

Answer: _____

END OF EXAMINATION PAPER